

Large Language Models are Zero-Shot Reasoners (Kojima et al., 2022)

Julie Weeds

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Kojima et al. (2022) demonstrate the effectiveness of prompt engineering for zero-shot reasoning. Read the paper and then answer the questions below:

1. Explain the terms: zero shot learning, few shot learning and in-context learning?
2. Why are mathematical reasoning problems potentially challenging for LLMs?
3. Give an example of how a model can be encouraged to think “step-by-step” using a few-shot prompting paradigm.
4. Give an example of how a model can be encouraged to think “step-by-step” using a zero-shot prompting paradigm.
5. Outline the two stages of prompting proposed by the authors?
6. What are the four categories of reasoning tasks considered in the experiments? Give an example of each kind of problem?
7. What baselines do the authors compare their zero-shot-CoT approach to? How do the authors ensure determinism of the methods?
8. How do the authors carry out answer cleansing?
9. How does zero-shot-CoT compare to the baselines at the different tasks? What other factors are there that affect performance?

References

Takeshi Kojima, Shixiang Shane Gu, Machel Reid, Yutaka Matsuo, and Yusuke Iwasawa. 2022. Large language models are zero-shot reasoners. In *Proceedings of 36th Conference on Neural Information Processing Systems (NeurIPS 2022)*.